THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** No	$1(-3)+(2)(2)=1$, not equal to 2.
2	**(d)** 4	Put x=0: $y=4$.
3	**(d)** -9	$\frac{-2-(7)}{5-(4)}=-9$.
4	**(b)** 1	Rewrite as $y=x+\ldots$; slope $=-\frac{a}{b}$.
5	**(c)** 8	$k(5)+(2)=42$ gives $k=8$.
6	**(c)** $(6,-4)$	$8(6)+9(-4)=12$
7	**(d)** -3	$y=mx+c$ with $y=-2$ at $x=-1$: $c=-2-(-1)(-1)$.
8	**(b)** 0	A horizontal line has slope 0; a vertical line has no (undefined) slope.
9	**(d)** Yes	$6(5)+(4)(-6)=6$, equal to 6.
10	**(b)** $\frac{59}{5}$	Put y=0: $x=\frac{59}{5}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$y=0$: $4x=36$ gives $x=9$; point $(9,0)$ (1); $x=0$: $6y=36$ gives $y=6$; point $(0,6)$ (1).	1+1
12	Substitute: $k(-3)+6(-6)=-57$; $-3k=-21$ (1); $k=7$ (1).	1+1
13	$5y=-4x+38$; $y=-\frac{4}{5}x+\frac{38}{5}$ (1); Slope $=-\frac{4}{5}$, y-intercept $=\frac{38}{5}$ (1).	1+1
14	$y=0$: $4x=4$ gives $x=1$; point $(1,0)$ (1); $x=0$: $-2y=4$ gives $y=-2$; point $(0,-2)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Slope of the first two points $=\frac{-3-(1)}{-1-(-3)}=-2$ (1); Slope between $(-1,-3)$ and $(1,k)$ must also be -2: $\frac{k-(-3)}{2}=-2$ (1); $k=-3-2\times 2=-7$ (1).	1+1+1
16	Use $y=mx+c$ (1); $-2=5(0)+c$ gives $c=-2$ (1); $y=5x-2$ (1).	1+1+1
17	Slope $=\frac{y_2-y_1}{x_2-x_1}=\frac{0-(1)}{-6-(-8)}$ (1); $=-\frac{1}{2}$ (2).	1+2
	OR: Slope of the first two points $=\frac{1-(-2)}{-2-(-5)}=1$ (1); Slope between $(-2,1)$ and $(1,k)$ must also be 1: $\frac{k-(1)}{3}=1$ (1); $k=1+1\times 3=4$ (1).	1+1+1
18	Use $y=mx+c$ (1); $0=6(-3)+c$ gives $c=18$ (1); $y=6x+18$ (1).	1+1+1
	OR: Slope $=\frac{y_2-y_1}{x_2-x_1}=\frac{8-(-7)}{-8-(-2)}$ (1); $=-\frac{5}{2}$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rewrite as $y=-2x+3$; table: $x=-1\Rightarrow y=5$, $x=0\Rightarrow y=3$, $x=1\Rightarrow y=1$, $x=2\Rightarrow y=-1$ (1); Plot the four points; they lie on one straight line (1); Slope $=-2$ (1); y-intercept $=3$, i.e. the point $(0,3)$ (2).	1+1+1+2
	OR: $F=\frac{9}{5}\times 50+32=122$ (1); For $F=212$: $\frac{9}{5}C=180$, so $C=100$ (2); Slope $\frac{9}{5}$: each 1°C rise raises F by 1.8° (2).	1+2+2
20	Rewrite as $y=x+2$; table: $x=-1\Rightarrow y=1$, $x=0\Rightarrow y=2$, $x=1\Rightarrow y=3$, $x=2\Rightarrow y=4$ (1); Plot the four points; they lie on one straight line (1); Slope $=1$ (1); y-intercept $=2$, i.e. the point $(0,2)$ (2).	1+1+1+2
	OR: $F=\frac{9}{5}\times 40+32=104$ (1); For $F=212$: $\frac{9}{5}C=180$, so $C=100$ (2); Slope $\frac{9}{5}$: each 1°C rise raises F by 1.8° (2).	1+2+2

Mistakes to avoid:

- Finding the x-intercept by putting $x=0$ (that is the y-intercept); for the x-intercept put $y=0$.
- Writing the slope of $ax+by=c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.

- Saying a linear equation in two variables has one solution; it has infinitely many.
- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as (y, x) instead of (x, y).
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $-\frac{1}{5}$	$\frac{5-(4)}{-7-(-2)} = -\frac{1}{5}$.
2	**(c)** $-\frac{5}{6}$	Rewrite as $y = -\frac{5}{6}x + \ldots$; slope $= -\frac{a}{b}$.
3	**(d)** 4	$k(4) + 4(3) = 28$ gives $k = 4$.
4	**(b)** $(-4, -2)$	$5(-4) + 4(-2) = -28$
5	**(b)** -7	$y = mx + c$ with $y = -3$ at $x = -2$: $c = -3 - (-2)(-2)$.
6	**(a)** undefined	A horizontal line has slope 0; a vertical line has no (undefined) slope.
7	**(c)** No	$4(-2) + (4)(6) = 16$, not equal to 4.
8	**(c)** $\frac{48}{5}$	Put y=0: $x = \frac{48}{5}$.
9	**(a)** $-\frac{2}{13}$	$\frac{8-(6)}{-7-(6)} = -\frac{2}{13}$.
10	**(a)** 1	Rewrite as $y = x + \ldots$; slope $= -\frac{a}{b}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute: $k(2) + 6(4) = 38$; $2k = 14$ (1); $k = 7$ (1).	1+1
12	$2y = -5x + 22$; $y = -\frac{5}{2}x + 11$ (1); Slope $= -\frac{5}{2}$, y-intercept $= 11$ (1).	1+1
13	$y = 0$: $7x = 42$ gives $x = 6$; point $(6, 0)$ (1); $x = 0$: $7y = 42$ gives $y = 6$; point $(0, 6)$ (1).	1+1
14	Substitute: $k(6) + 5(3) = 57$; $6k = 42$ (1); $k = 7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Slope of the first two points $= \frac{-5-(-1)}{5-(3)} = -2$ (1); Slope between $(5, -5)$ and $(7, k)$ must also be -2: $\frac{k-(-5)}{2} = -2$ (1); $k = -5 - 2 \times 2 = -9$ (1).	1+1+1
16	Use $y = mx + c$ (1); $6 = 1(-3) + c$ gives $c = 9$ (1); $y = x + 9$ (1).	1+1+1
17	Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{-6-(0)}{-9-(3)}$ (1); $= \frac{1}{2}$ (2).	1+2
	OR: Slope of the first two points $= \frac{-3-(3)}{4-(1)} = -2$ (1); Slope between $(4, -3)$ and $(7, k)$ must also be -2: $\frac{k-(-3)}{3} = -2$ (1); $k = -3 - 2 \times 3 = -9$ (1).	1+1+1
18	Use $y = mx + c$ (1); $-4 = -2(4) + c$ gives $c = 4$ (1); $y = -2x + 4$ (1).	1+1+1
	OR: Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{7-(-1)}{-6-(-4)}$ (1); $= -4$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rewrite as $y = -x + 0$; table: $x = -1 \Rightarrow y = 1$, $x = 0 \Rightarrow y = 0$, $x = 1 \Rightarrow y = -1$, $x = 2 \Rightarrow y = -2$ (1); Plot the four points; they lie on one straight line (1); Slope $= -1$ (1); y-intercept $= 0$, i.e. the point $(0, 0)$ (2).	1+1+1+2
	OR: $F = \frac{9}{5} \times 30 + 32 = 86$ (1); For $F = 212$: $\frac{9}{5}C = 180$, so $C = 100$ (2); Slope $\frac{9}{5}$: each 1°C rise raises F by 1.8° (2).	1+2+2
20	Rewrite as $y = 4x - 2$; table: $x = -1 \Rightarrow y = -6$, $x = 0 \Rightarrow y = -2$, $x = 1 \Rightarrow y = 2$, $x = 2 \Rightarrow y = 6$ (1); Plot the four points; they lie on one straight line (1); Slope $= 4$ (1); y-intercept $= -2$, i.e. the point $(0, -2)$ (2).	1+1+1+2
	OR: $F = \frac{9}{5} \times 45 + 32 = 113$ (1); For $F = 212$: $\frac{9}{5}C = 180$, so $C = 100$ (2); Slope $\frac{9}{5}$: each 1°C rise raises F by 1.8° (2).	1+2+2

Mistakes to avoid:

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.

- Saying a linear equation in two variables has one solution; it has infinitely many.
- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as (y, x) instead of (x, y).
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 3	$k(5)+6(2)=27$ gives $k=3$.
2	**(a)** $(6,-5)$	$5(6)+7(-5)=-5$
3	**(b)** -12	$y=mx+c$ with $y=-3$ at $x=3$: $c=-3-(3)(3)$.
4	**(c)** undefined	A horizontal line has slope 0; a vertical line has no (undefined) slope.
5	**(c)** Yes	$6(4)+(4)(-5)=4$, equal to 4.
6	**(c)** $\frac{26}{3}$	Put y=0: $x=\frac{26}{3}$.
7	**(c)** -3	$\frac{7-(-2)}{-1-(2)}=-3$.
8	**(a)** $-\frac{1}{2}$	Rewrite as $y=-\frac{1}{2}x+\ldots$; slope $=-\frac{a}{b}$.
9	**(c)** 7	$k(1)+4(4)=23$ gives $k=7$.
10	**(d)** $(5,7)$	$6(5)+7(7)=79$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2y=-7x+34$; $y=-\frac{7}{2}x+17$ (1); Slope $=-\frac{7}{2}$, y-intercept $=17$ (1).	1+1
12	$y=0$: $2x=32$ gives $x=16$; point $(16,0)$ (1); $x=0$: $4y=32$ gives $y=8$; point $(0,8)$ (1).	1+1
13	Substitute: $k(5)+4(0)=20$; $5k=20$ (1); $k=4$ (1).	1+1
14	$-5y=-3x+8$; $y=\frac{3}{5}x-\frac{8}{5}$ (1); Slope $=\frac{3}{5}$, y-intercept $=-\frac{8}{5}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Slope of the first two points $=\frac{5-(3)}{2-(0)}=1$ (1); Slope between $(2,5)$ and $(4,k)$ must also be 1: $\frac{k-(5)}{2}=1$ (1); $k=5+1\times 2=7$ (1).	1+1+1
16	Use $y=mx+c$ (1); $-3=-3(6)+c$ gives $c=15$ (1); $y=-3x+15$ (1).	1+1+1
17	Slope $=\frac{y_2-y_1}{x_2-x_1}=\frac{-5-(4)}{7-(5)}$ (1); $=-\frac{9}{2}$ (2).	1+2
	OR: Slope of the first two points $=\frac{13-(4)}{1-(-2)}=3$ (1); Slope between $(1,13)$ and $(4,k)$ must also be 3: $\frac{k-(13)}{3}=3$ (1); $k=13+3\times 3=22$ (1).	1+1+1
18	Use $y=mx+c$ (1); $-4=-5(6)+c$ gives $c=26$ (1); $y=-5x+26$ (1).	1+1+1
	OR: Slope $=\frac{y_2-y_1}{x_2-x_1}=\frac{-2-(9)}{4-(5)}$ (1); $=11$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rewrite as $y=4x+0$; table: $x=-1\Rightarrow y=-4$, $x=0\Rightarrow y=0$, $x=1\Rightarrow y=4$, $x=2\Rightarrow y=8$ (1); Plot the four points; they lie on one straight line (1); Slope $=4$ (1); y-intercept $=0$, i.e. the point $(0,0)$ (2).	1+1+1+2
	OR: $F=\frac{9}{5}\times 20+32=68$ (1); For $F=212$: $\frac{9}{5}C=180$, so $C=100$ (2); Slope $\frac{9}{5}$: each 1°C rise raises F by 1.8° (2).	1+2+2
20	Rewrite as $y=2x-4$; table: $x=-1\Rightarrow y=-6$, $x=0\Rightarrow y=-4$, $x=1\Rightarrow y=-2$, $x=2\Rightarrow y=0$ (1); Plot the four points; they lie on one straight line (1); Slope $=2$ (1); y-intercept $=-4$, i.e. the point $(0,-4)$ (2).	1+1+1+2
	OR: $F=\frac{9}{5}\times 55+32=131$ (1); For $F=212$: $\frac{9}{5}C=180$, so $C=100$ (2); Slope $\frac{9}{5}$: each 1°C rise raises F by 1.8° (2).	1+2+2

Mistakes to avoid:

- Finding the x-intercept by putting $x=0$ (that is the y-intercept); for the x-intercept put $y=0$.
- Writing the slope of $ax+by=c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
- Saying a linear equation in two variables has one solution; it has infinitely many.

- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as (y, x) instead of (x, y).
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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